

Physics of cutting — Solution

Y26 (2026) — English translation

A1

0.80 pts

Having written the law of conservation of energy for the system, obtain the equation for the connection between θ , R , μ , w , F_C and N .

Consider the displacement of the knife by dr in the cutting direction. The cut does not stretch, so its relative displacement along the wedge is also equal to dr . Let's write down the law of conservation of energy for the sawed-off knife system:

$$F_C dr = R w dr + \mu N dr$$

From here we get:

Answer:

$$F_C = \mu N + R w$$

A2

0.80 pts

Define F_C . Express your answer using R , μ , θ , w .

Let's write down Newton's second law for a wedge in projection onto the horizontal axis:

$$F_C = \mu N \cos \theta + N \sin \theta$$

Combining this expression with the result of the previous paragraph, we get:

Answer:

$$F_C = \frac{R w}{1 - \frac{\mu}{\mu \cos \theta + \sin \theta}}$$

A3

1.00 pts

Construct a qualitative plot of $F_C(\theta)$ against $\mu = 0.3$ in the range $\theta \in [0; \pi/2]$. Specify special points and their coordinates. Express F_C in units of wR .

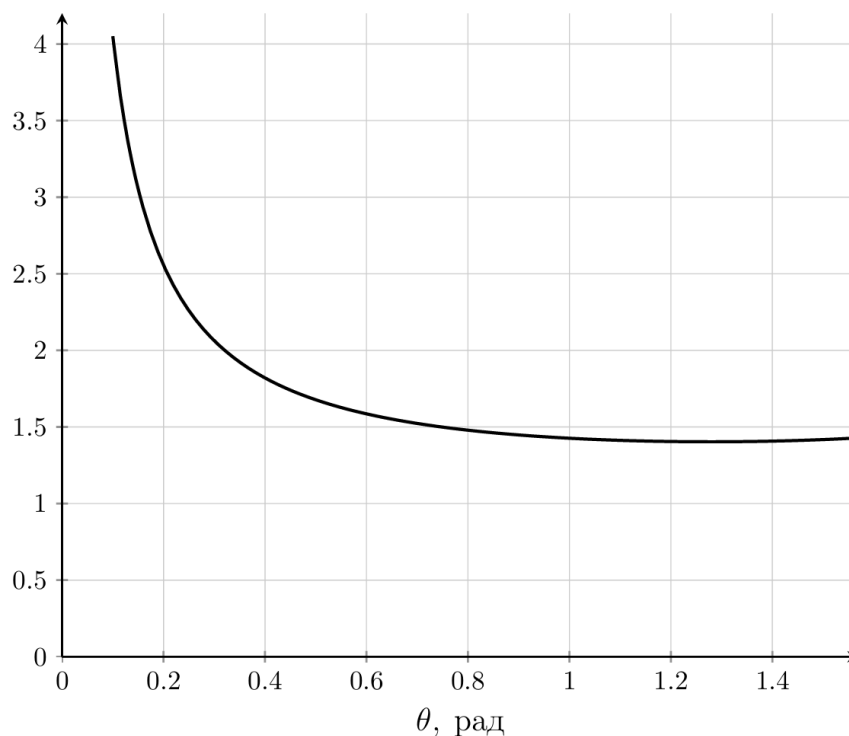
Let's build a graph:

This dependence has an extremum:

$$\cos \theta - \mu \sin \theta = 0, \quad \theta = \arctg \frac{1}{\mu} \approx 73^\circ$$

The minimum value of the force F_C is expressed as:

$$F_C = \frac{R w}{1 - \frac{\mu}{\sqrt{\mu^2 + 1}}} \approx 1.40 R w$$



A4

0.20 pts

Determine F_C and F_T , neglecting friction. Express your answers using R , θ , w .

Simplifying the answer from A2, we get:

In the absence of friction, the total force acting on the wedge is directed at an angle θ to the vertical, therefore:

Answer:

$$F_C = Rw$$

B1

0.20 pts

Determine the total horizontal force F acting on the knife, neglecting friction. Express your answer using R , θ , w , ψ .

There is no need to apply force to move parallel to the cutting line, therefore:

Answer:

$$F = Rw$$

B2

1.50 pts

Determine the total horizontal force F acting on the knife from the side of the chips. Express your answer in terms of R , w , ψ , θ , μ .

Let us introduce the projections $F_{\parallel}$ and $F_{\perp}$. According to the Coulomb-Amonton law, the friction force acting on the chips is equal to μN in absolute value and is directed against the speed of movement of the chips relative to the wedge. Let's write down Newton's second law:

$$F_{\perp} = N(\sin \theta + \mu \cos \theta \cos \psi)$$

In the projection onto the perpendicular horizontal axis, the equilibrium condition is as follows:

$$F_{\parallel} = \mu N \sin \psi$$

Let's write down the ESE, for this we equate the power of the force F to the power of energy loss in the

system due to friction:

$$\mu N v + R w v \cos \psi = F_{\perp} v \cos \psi + F_{\parallel} v \sin \psi$$

Express N , combining the results:

$$N = \frac{R w}{(\sin \theta + \mu \cos \theta \cos \psi) - \mu \cos \psi}$$

Express F like $F = \sqrt{F_{\parallel}^2 + F_{\perp}^2}$:

Answer:

$$F = \frac{R w}{(\sin \theta + \mu \cos \theta \cos \psi) - \mu \cos \psi} \sqrt{\mu^2 \sin^2 \psi + (\sin \theta + \mu \cos \theta \cos \psi)^2}$$

C1

0.30 pts

From geometry, determine t_c . Express your answer in terms of t , angles φ and α .

The chip length remains constant, hence:

$$\frac{t}{\sin \varphi} = \frac{t_c}{\cos(\varphi - \alpha)}$$

Let us express t_c :

Answer:

$$t_c = t \frac{\cos(\varphi - \alpha)}{\sin \varphi}$$

C2

0.30 pts

Find at what angle φ the chip thickness t_c is equal to t .

Let's equate the sines of the angles:

$$\sin \varphi = \cos(\varphi - \alpha) = \sin(\pi/2 - \varphi + \alpha)$$

From here:

Answer:

$$\varphi = \pi/4 + \alpha/2$$

C3

1.00 pts

Define γ . Express your answer using φ , α .

First solution

Consider the velocity triangle:

From here, according to the theorem of sines:

$$\frac{|\vec{v}_2 - \vec{v}_1|}{|\vec{v}_1|} = \frac{\cos \alpha}{\cos(\varphi - \alpha)}$$

Let us express the answer:

Second solution

From the metal incompressibility equation:

$$v_1 t = v_2 t_c$$

Let us express the answer:

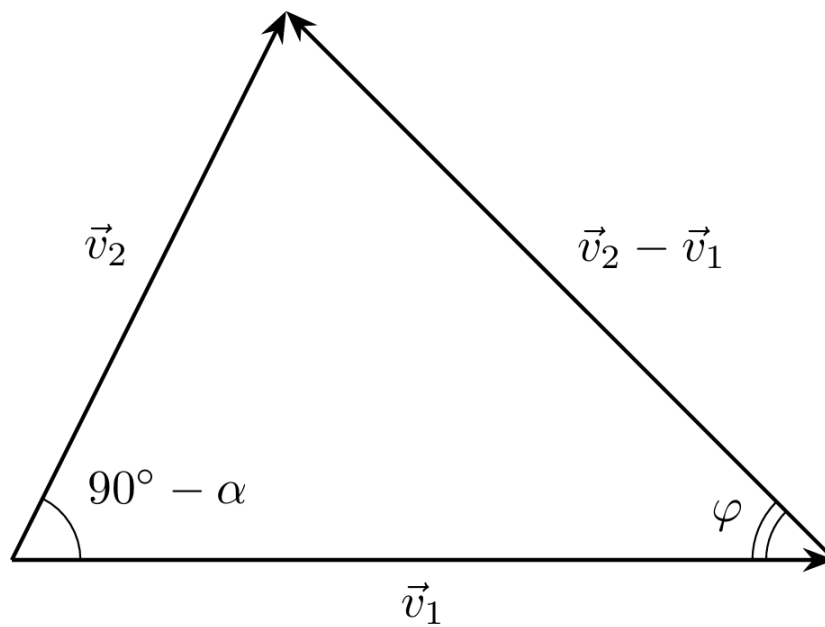
$$\gamma = \frac{\sqrt{1 + \left(\frac{\sin \varphi}{\cos(\varphi - \alpha)} \right)^2 - 2 \frac{\sin \varphi}{\cos(\varphi - \alpha)} \sin \alpha}}{\sin \varphi} = \frac{\sqrt{\cos^2 \varphi \cos^2 \alpha + \sin^2 \varphi \sin^2 \alpha + 2 \sin \varphi \sin \alpha \cos \varphi \cos \alpha + \sin^2 \varphi - 2 \sin \varphi \sin \alpha}}{\sin \varphi \cos(\varphi - \alpha)}$$

By simple transformations, the answer is simplified to:

$$\gamma = \frac{\cos \alpha}{\cos(\varphi - \alpha) \sin \varphi}$$

Answer:

$$\gamma = \frac{|\vec{v}_2 - \vec{v}_1|}{|\vec{v}_1|} = \frac{\cos \alpha}{\cos(\varphi - \alpha) \sin \varphi}$$



C4

1.20 pts

Define F_C . Express your answer as $w, k, t, \varphi, \alpha, \beta, R, \gamma$.

Let us associate the friction force acting on the cut with F_C , for this we write down the equality of the horizontal forces acting on the wedge:

$$F_C = N(\mu \sin \alpha + \cos \alpha)$$

We express the friction sieve F_t :

$$F_t = \mu N = \frac{\mu F_C}{\mu \sin \alpha + \cos \alpha} = \frac{F_C \sin \beta}{\cos(\beta - \alpha)}$$

We write the law of conservation of energy:

$$F_C dr = R w dr + F_t dr \frac{|v_2|}{|v_1|} + k \gamma w t dr$$

From the theorem of sines:

$$\frac{v_2}{v_1} = \frac{\sin \varphi}{\cos(\varphi - \alpha)}$$

Combining the results, we obtain:

Answer:

$$F_C = \frac{w(k\gamma t + R)}{1 - \frac{\sin \beta \sin \varphi}{\cos(\varphi - \alpha) \cos(\beta - \alpha)}}$$

C5

0.10 pts

Write down the expression F_C for metals. Express your answer as w, k, t, φ, α and β .

Let's substitute the value γ and neglect R :

Answer:

$$F_C = \frac{wkt \cos \alpha \cos(\beta - \alpha)}{\sin \varphi (\cos(\varphi - \alpha) \cos(\beta - \alpha) - \sin \beta \sin \varphi)}$$

C6

1.00 pts

Calculate the value of the angle φ at which the force F_C is minimal. Express your answer using β and α . Also determine the force F_C according to Merchant's theory. Express your answer using w, k, t, β and α .

Let's find the maximum denominator. To do this, we equate the derivative of the denominator to zero:

$$\cos \varphi \cos(\varphi - \alpha) - \sin \varphi \sin(\varphi - \alpha) - \frac{2 \sin \beta \sin \varphi \cos \varphi}{\cos(\beta - \alpha)} = 0$$

From here:

$$\cos(2\varphi - \alpha) \cos(\beta - \alpha) = \sin 2\varphi \sin \beta$$

We transform the resulting expressions using the formulas for half-sum and half-difference of cosines:

$$\frac{\cos(2\varphi + \beta - 2\alpha) + \cos(2\varphi - \beta)}{2} = \frac{\cos(2\varphi - \beta) - \cos(2\varphi + \beta)}{2}$$

From here we get the equation:

$$\pi - 2\varphi - \beta = 2\varphi + \beta - 2\alpha$$

Optimal angle:

Let's substitute the value φ and determine the minimum possible force:

Answer:

$$\varphi = \frac{\pi}{4} + \frac{\alpha - \beta}{2}$$

D1

0.80 pts

Determine the force F acting on the blade during slow cutting. Express your answer using $R, t, w, d\Lambda/da$.

Let's write down the law of conservation of energy:

$$Fda = Rwd a - d\Lambda$$

From here:

Answer:

$$F = R w - d\Lambda/da$$

D2

0.80 pts

At what minimum initial deformation ε can the cut propagate spontaneously, without expending external work? Express your answer in terms of R, E and t .

Spontaneous spread of the cut is possible at $F \leq 0$. Then:

$$R w - d\Lambda/da \leq 0$$

Surface energy density of the material:

$$d\Lambda/dS = \frac{1}{2} E \varepsilon^2 w$$

Let us express the area of the free triangular region:

$$S = ta$$

From here:

$$d\Lambda/da = \frac{1}{2}E\varepsilon^2wt$$

Then:

Answer:

$$\varepsilon_{\min} = \sqrt{\frac{2R}{Et}}$$